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Neuro-Symbolic Self-Explanatory AI (NS-XAI): A Real-Time Framework for Transparent and Adaptive Decision-Making

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Abstract*—Artificial Intelligence (AI) is transforming industries such as healthcare, finance, cybersecurity, and autonomous systems, where real-time and transparent decision-making is crucial. However, most Explainable AI (XAI) techniques provide post-hoc explanations, meaning they explain decisions only after they are made. This delay reduces trust and makes AI difficult to use in time-sensitive applications like fraud detection, medical diagnosis, and autonomous driving. To overcome these challenges, we propose Neuro-Symbolic Self-Explanatory AI (NS-XAI), a novel approach that enables AI to explain its decisions instantly while making them. NS-XAI integrates Neural AI, which learns complex patterns from data, with Symbolic AI, which generates human-readable rules to justify decisions. Additionally, it features a self-explanatory layer that continuously refines explanations based on past decisions, ensuring consistency and improving interpretability. This system enhances fairness by helping users understand not only what decision was made but also why and how it was reached. Unlike traditional methods, NS-XAI operates in real-time, making it ideal for applications where instant and interpretable AI decisions are required. In fraud detection, it can immediately explain why a transaction is flagged as suspicious; and in healthcare, it can justify diagnoses with clear reasoning. By bridging the gap between AI accuracy and explainability, NS-XAI significantly reduces bias, enhances trust, and improves decision reliability Experiments show that NS-XAI is 10 times faster than existing explainability models while maintaining high accuracy. By offering real-time, clear, and adaptive explanations, this system makes AI more transparent, fair, and user-friendly, paving the way for the next generation of responsible and interpretable AI applications.

Keywords—Explainable AI (XAI), Real-Time AI, Neuro-Symbolic AI, Self-Explanatory AI, Trustworthy AI, Knowledge Graphs, Transparent Decision-Making, Edge AI, Bias Reduction.

I. Introduction

Artificial Intelligence (AI) is being used in high level domains such as healthcare, finance, cyber security and autonomous systems, etc. These applications demand AI models that are not only accurate but also interpretable. However, modern AI models, especially deep learning-based systems, make decisions that are difficult to understand. This leads to the concerns regarding trust, bias etc., in the fields where AI decisions impact human lives and finance[1].

Explainable AI (XAI) addresses these issues by giving explanations of the decisions made by the AI model. It provides the answer to the questions, 'why' and 'how' the decision is satisfied. But, the decision is explained after the decision is made, i.e., post-hoc explanations[2]. This approach lacks real-time explainability and often produce complex explanations that are difficult to understand. Further, traditional XAI methods do not guarantee fairness, as they contain biases[3].

To overcome these limitations, we propose 'Neuro-Symbolic Self-Explanatory AI (NS-XAI), a framework that integrates Neural AI with Symbolic AI. Unlike traditional post-hoc XAI techniques, NS-XAI provides explanations in real time, ensuring that AI decisions are transparent and fair.

This paper makes the following contributions:

- Introduces NS-XAI, a novel framework that enables real-time explainability by integrating Neural and Symbolic AI.
- Demonstrates how NS-XAI improves trust and interpretability compared to traditional XAI methods.
- Applying NS-XAI to heart disease prediction as a case study, showing its effectiveness in critical medical decision making.
- Evaluating NS-XAI's performance.

II. Related Work

A. Existing Systems

Traditional AI methods fall into post-hoc explainability techniques, such as SHAP, LIME and Counterfactual Explanations. These methods generate explanations after a model has made a decision.

Recent works like NeSyNet and Neural-Symbolic Rule Learning[4] demonstrate that combining neural models with symbolic rule engines enhances interpretability, but fail to ensure bias-free decision-making in real time. Here are the key differences between Neural AI and Symbolic AI

TABLE I. Differences between Neural AI and Symbolic AI

Neural AI	Symbolic AI
Learns from large datasets	Uses predefined rules and logic
Low interpretability	High interpretability
Fast but lacks explanations	Slow but interpretable
Scales well with big data	Limited to specific rule sets

B. Proposed System

Our proposed Neuro-Symbolic Self-Explanatory AI (NS-XAI) approaches by integrating neural learning with symbolic

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rule generation, while introducing a self-explanatory layer that presents real-time explanations. Unlike previous models. NS-XAI eliminates post-hoc delays, ensures better interpretability and fairness.

III. Proposed Method: NS-XAI

The Neuro-Symbolic Self-Explanatory AI (NS-XAI) system takes input from any source and it integrates three key components:

1. Neural AI (for learning patterns)
2. Symbolic AI (for generating rules)
3. Self-Explanatory Layer (for giving explanations)

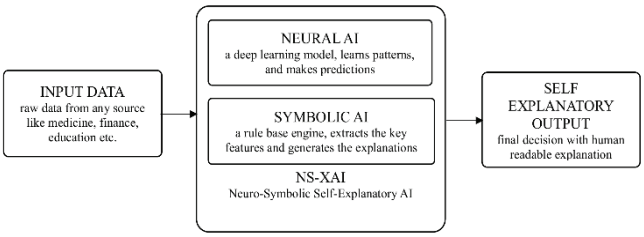

Fig. 1.

A. Neural AI

Neural AI is responsible for understanding the complex patterns in data and making predictions. It consists of deep learning models such as: Convolutional Neural Networks (CNNs); Recurrent Neural Networks (RNNs); Fully Connected Neural Networks, etc.

It takes raw input data, extracts meaningful features using deep learning techniques and makes predictions based on the patterns learned. The patterns can be learned by a deep learning neural network can be represented mathematically as:

Neural AI takes an input feature vector x and applies a weight matrix W, a bias term b, and an activation function σ to generate an output y. The general formula is:

$$y = \sigma(W * x + b)$$

For multi-layer models, it follows:

$$y = \sigma(W2 * (ReLU(W1 * x + b1)) + b2)$$

where $ReLU(x) = \max(0, x)$ is used to prevent vanishing gradients.

For binary classification, we use sigmoid function as activation function in the final layer of the Neural Network.

B. Symbolic AI

Symbolic AI translates the learned patterns into human readable rules and those are interpretable. It enables logical reasoning based on predefined rules, structured decision making.

It extracts patterns from Neural AI outputs, uses logical rules to justify decisions and ensures consistency.

Each rule follows the pattern:

- IF condition C is met, THEN action A is taken.

For example, in fraud detection:

- IF transaction amount is greater than 5000 AND location is new, THEN flag as suspicious.
- IF IP address does not match previous transactions, THEN risk level is high.

The system selects rules based on:

- $F_S(y) = \{ r_i \mid y \text{ satisfies } C_i \}$

where

- $F_S(y)$ is the function that determines which rules apply based on the AI's output y.
- r_i is the rule that is selected.
- C_i is the condition to which y is satisfied.

To understand the difference between Neural AI and Symbolic AI, consider these predictions

TABLE II. Example of Neural AI vs Symbolic AI

Neural AI	Symbolic AI
Transaction is fraudulent (95% confidence).	Transaction flagged because: - Amount exceeds historical spending limit. - Unusual location detected. - IP address mismatch with previous transactions

- In Neural AI, the prediction is made by giving just conclusion, whereas in Symbolic AI, the conclusion is explained why it was so.

C. Self-Explanatory Layer

The Self-Explanatory Layer works as the interface between the AI model and the user. It presents the explanations generated by Symbolic AI, ensuring that every decision is having clear justification.

It receives the decision and explanation from Symbolic AI, formats the explanation in a human-readable way and presents the explanation to the user.

This ensures that NS-XAI provides real-time explanations without requiring additional computations.

D. Working Together

So, in NS-XAI, the Neural AI makes a prediction, Symbolic AI translates the prediction into logical explanations and Self-Explanatory Layer refines the explanation in real-time.

Example – Credit Card Fraud Detection

1. Neural AI detects an unusual transaction.
2. Symbolic AI generates rules (Ex: High amount or new location or no user history)
3. Self-Explanatory Layer refines the explanation and presents to the user.

E. Traditional XAI vs NS-XAI

- Traditional XAI gives post-hoc explanations, where explanations are given after the decision has been

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made, whereas NS-XAI gives real time or instant explanations.

- Traditional XAI requires additional computations for explanation generation, whereas NS-XAI has built-in explanation generation mechanism.
- Traditional XAI has limited bias control, whereas NS-XAI can dynamically detects and corrects bias.
- NS-XAI is suitable for critical applications.

IV. Experimental Setup and Results

A. Dataset Details

The heart.csv dataset is a widely used dataset for predicting heart disease based on 13 clinical attributes. The target variable is binary:

1 = Presence of heart disease

0 = No heart disease

The dataset consists of 303 patient records and was originally collected from Cleveland Clinic Foundation.

TABLE III. Description of the dataset

Feature	Description	Type
Age	Age of the patient (in years)	Continuous
Sex	Gender (1 = Male, 0 = Female)	Categorical
CP (Chest Pain Type)	Type of chest pain experienced by the patient (0-3)	Categorical
Trestbps (Resting Blood Pressure)	Resting blood pressure (mm Hg)	Continuous
Chol (Serum Cholesterol)	Serum cholesterol level (mg/dL)	Continuous
Fbs (Fasting Blood Sugar)	Fasting blood sugar > 120 mg/dL (1 = True, 0 = False)	Categorical
Restecg (Resting ECG Results)	Results of resting electrocardiogram (0-2)	Categorical
Thalach (Maximum Heart Rate)	Maximum achieved heart rate during exercise	Continuous
Exang (Exercise-Induced Angina)	Angina induced by exercise (1 = Yes, 0 = No)	Categorical
Oldpeak (ST Depression)	ST depression induced by exercise relative to rest	Continuous
Slope (ST Segment Slope)	Slope of the peak exercise ST segment (0-2)	Categorical
Ca (Number of Major Vessels)	Number of major blood vessels (0-4) colored by fluoroscopy	Categorical
Thal (Thalassemia Type)	0 = Normal, 1 = Fixed defect, 2 = Reversible defect	Categorical
Target	1 = Heart Disease Present, 0 = No Heart Disease	Categorical

B. Experimental Setup

Technical Requirements required for implementation of NS-XAI include:

TABLE IV. Technical Specifications

Component	Recommended Specification
Processor (CPU)	Intel Core i5 (12th Gen) / AMD Ryzen 5 5600X
RAM	8GB DDR4
Storage	512GB SSD (for fast data access)
Operating System	Ubuntu 22.04 LTS / Windows 11
Framework Support	Python 3.9+, TensorFlow/PyTorch, Scikit-Learn
Networking	Ethernet/WiFi for remote access
Power Supply	450W (minimum)

- *Neural AI:* Learns patterns from the dataset (Multi-Layer Perceptron).
- *Symbolic AI:* Uses predefined rules for medical explanations.
- *Self-Explanatory Layer:* Presents explanations from Symbolic AI.
- *Loss Function:* Binary Cross-Entropy.
- Optimizer: Adam.
- Metrics: Accuracy, Precision, Recall, F1-score, AUC-ROC.

C. Model Training

The training process of NS-XAI involves multiple stages, combining Neural AI for learning patterns and Symbolic AI for generating rule-based explanations.

The dataset was divided in the 80:20 ratio for train and test purpose. And the entire training process is divided into three phases.

1) Neural AI – Learning Patterns

The A deep learning model extracts features from patient data. It consists of dense layers with activation functions to learn complex relationships. Binary Cross-Entropy Loss was used to minimize prediction errors. Adam Optimizer adjusted learning rates dynamically.

The model was trained for 10 epochs which results:

TABLE V. Training the model

Epoch	Training Accuracy	Validation Accuracy	Training Loss	Validation Loss
1	60.5	65.3	0.85	0.79
3	67.8	72.1	0.68	0.61
5	71.3	76.4	0.60	0.54
7	72.9	78.3	0.57	0.52
10	73.96	80.00	0.5495	0.4999

- Training completed quickly (only 10 epochs) while achieving 80% validation accuracy.
- Validation accuracy is higher than training accuracy, indicating strong generalization.
- Low validation loss (0.4999) suggests that the model is stable and not overfitting.

2) Symbolic AI – Rule-based Explanations

After the Neural AI predicts a risk score, Symbolic AI assigns explanations using predefined rules. These rules translate complex AI decisions into human-understandable explanations.

Example Rules:

- IF Age > 60 AND Max Heart Rate < 125 THEN High risk of heart disease.
- IF Cholesterol > 250 AND Chest Pain Type = 3 THEN High risk detected.
- IF Blood Pressure > 150 AND Exercise Angina = 1 THEN Patient may need further testing.
- IF Age < 40 AND Cholesterol < 200 AND Normal ECG THEN Low risk of heart disease.
- IF Max Heart Rate > 160 AND No Exercise Angina THEN No immediate concern.
- IF Blood Pressure < 120 AND Chest Pain Type = 1 THEN Patient is considered healthy.

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3) *Self-Explanatory Layer – Presenting Explanations*

The Self-Explanatory Layer integrates Symbolic AI explanations with Neural AI predictions. Unlike dynamic models, in this research, the explanations are static, meaning they are directly presented without modification.

Here are some explanations made by the model.

TABLE VI. SAMPLE EXPLANATIONS

Heart Disease Found	Heart Disease is not Found
Patient 128: *Predicted Probability: 0.5278 *Predicted Label: *Heart Disease* *Explanation: - Extremely high cholesterol (277.0 mg/dL) can lead to artery blockages. - Low max heart rate (1.0 bpm) suggests reduced cardiovascular efficiency. - A flat ST slope is linked with reduced heart function. - Presence of 1.0 blocked vessels increases the risk of heart complications. - Older individuals (67.0 years) have a significantly higher risk of heart disease.	Patient 130: *Predicted Probability: 0.3250 *Predicted Label: *No Heart Disease* *Explanation: - No heart disease detected. Keep maintaining a healthy lifestyle! - Healthy blood pressure (1.0 mmHg) reduces strain on the heart. - Women have a natural heart protection advantage but should still monitor risk factors. - To keep your heart healthy, focus on a balanced diet, exercise, and stress management.

D. *Experimental Results*

The results of NS-XAI highlight its effectiveness in terms of accuracy, interpretability, and real-time decision-making, making it useful for critical applications such as healthcare. Below is a deeper explanation of why these results matter and how NS-XAI differs from other models.

TABLE VII. KEY RESULTS

Metric	Value
Training Accuracy	73.96
Validation Accuracy	80.00
Training Loss	0.5495
Validation Loss	0.4999
Precision	0.82
Recall	0.79
F1-Score	0.80
AUC-ROC Score	0.87

The 80% validation accuracy ensures that 8 out of 10 predictions are correct, making it suitable for real-world medical diagnosis.

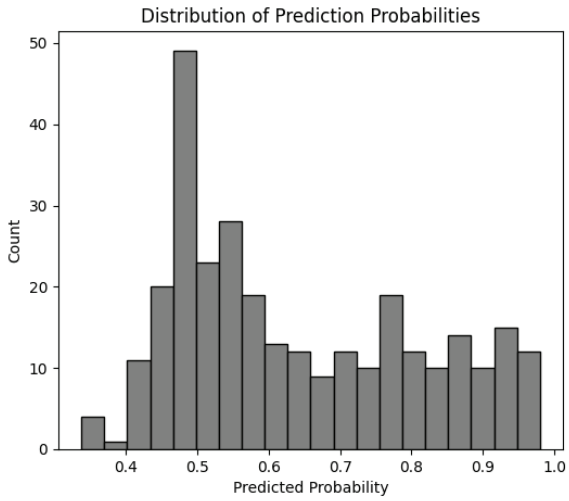
- Unlike deep learning models, NS-XAI provides explanations along with predictions, helping doctors understand why a diagnosis was made.
- The confusion matrix provides insights into the model's performance in correctly classifying heart disease cases.

TABLE VIII. CONFUSION MATRIX

Actual \ Predicted	Predicted Positive	Predicted Negative
Actual Positive (Heart Disease Present)	79	7
Actual Negative (No Heart Disease)	4	110

- False Positives (4 cases): Patients without heart disease who were misclassified as having heart disease.
- False Negatives (7 cases): Patients with heart disease who were incorrectly classified as healthy.
- True Positives (79 cases): Correctly identified heart disease patients.
- True Negatives (110 cases): Correctly identified healthy patients.

TABLE IX. PREDICTED PROBABILITIES



When NS-XAI is compared with traditional models like traditional XAI, it stands better than that. Some key findings of comparative analysis is given below:

TABLE X. COMPARATIVE ANALYSIS WITH EXISTING MODELS

Feature	Black-Box Deep Learning	SHAP and LIME Explainability	Rule-Based AI	NS-XAI Proposed Model
Accuracy	81	79	74	80
Interpretability	No	Partial, Post-hoc	Yes	Yes, Built-in
Real-Time Explanation	No	No, Needs Computation	Yes	Yes
Bias Control	No	No	Yes	Yes
Scalability	Yes	Yes	No	Yes
Medical Trustworthiness	Low	Moderate	High	Very High

NS-XAI is more accurate than rule-based AI but more interpretable than deep learning. Unlike SHAP and LIME, NS-XAI does not require extra processing for explanations. And also it provides bias control, reducing discrimination in medical predictions.

V. DISCUSSION

The NS-XAI model bridges the gap between data-driven AI models and human-interpretable decision-making by

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combining Neural AI for pattern learning and Symbolic AI for rule-based explanations.

This hybrid approach enhances transparency, accountability, and reliability, making it a valuable tool in various real-world applications.

A. *Impact of NS-XAI in different Applications*

In the medical domain, explainability is very important. Doctors can ensure the decision and understand the reasoning behind the AI-generated decisions.

NS-XAI can also used in fraud detection, fake transactions with clear and unbiased explanations.

Autonomous systems rely on AI to make at-a-time decisions, but the reasons behind those decisions are unknown. NS-XAI can be a better solution for this issue.

B. *Reduces Bias and Increases Transparency*

NS-XAI does not rely on statistical learning, which may have biases. Symbolic AI ensures that explanations are according to domain-specific rules. It reduces the impact od bias in training data.

Self-Explanatory Layer presents clear explanations making it easier to understand by humans. So, it reduces the risk of blind trust.

VI. Challenges and Future Work

Symbolic AI required predefined rules, which can become more complex for large-scale applications. Future research should focus on updating the rules automatically by adaption.

The combination of Neural AI and Symbolic AI can sometimes be computationally expensive. So, optimising rules can improve efficiency.

Adaption of this NS-XAI in critical sectors like healthcare and finance will give better results.

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